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Nitrogen-fixing bacteria such as *Azotobacter chroococcum* use the enzyme nitrogenase to produce ammonia from nitrogen gas in the air. *A. chroococcum* can use ammonium chloride as a direct source of ammonia. When a source of ammonia is not available this bacterium uses nitrogen fixation.

A scientist investigated the effect of an increase in the concentration of ammonium chloride on the activity of nitrogenase in this bacterium. He prepared several liquid medium cultures of the bacterium. Each liquid culture had the same volume. He grew each culture in a different concentration of ammonium chloride.

In each culture:

- he recorded the nitrogenase activity in arbitrary units
- he removed the bacteria and then recorded the concentration of ammonium chloride remaining in each liquid medium.

Table 2 shows the scientist's results.

Table 2

Concentration of ammonium chloride / $\mu\text{g cm}^{-3}$	Nitrogenase activity / arbitrary units	Concentration of ammonium chloride remaining in liquid medium / $\mu\text{g cm}^{-3}$
0	45	0
20	30	0
40	17	0
60	7	0
80	0	6
100	0	14
120	0	20

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Apart from temperature and pH, give **two** variables the scientist would have controlled when **preparing** the liquid medium cultures.

[2 marks]

1 _____

2 _____

[illegible]

